



Filtering Electroencephalogram Data Using R: A Study on Patients with Seizures

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Abstract

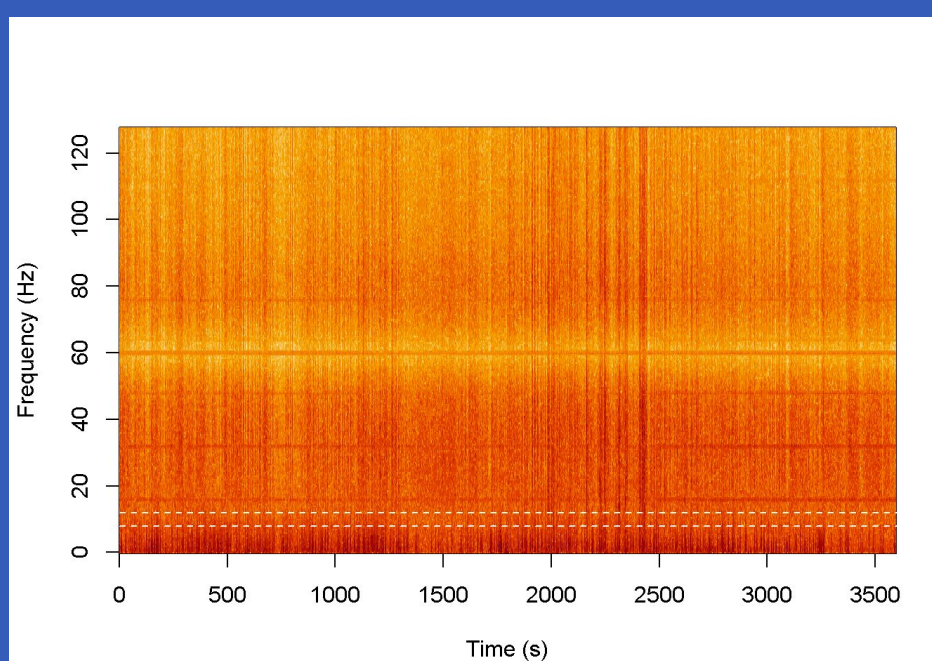
Seizures pose an imperative risk due to their frequency rising after the COVID-19 pandemic. EEG data can be used to determine the location and timing of seizures to prevent future ones from occurring. For data to be useful, it needs to be filtered to reduce interference and noise within biosignals and to prevent the number of false alarms in current tracking methods. This experiment tested three different filters through RStudio: a Butterworth bandpass filter tested in three different orders which emphasized signals in certain frequencies while diminishing signals in others; a moving average filter that smooths the data without specific cutoffs; and a zero-phase filter that smooths the noise while preserving original timing. These three functions were tested on synthetic sine waves generated at 5 Hz and 60 Hz to measure their effectiveness on the sample dataset.

Based on the results, the Butterworth bandpass filter is the most effective for real-life applications, such as wearable IoT devices used to track seizures when they first start to flare up in specific electrodes. Likewise, the zero-phase filter is preferred when analyzing entire signals and massive datasets, since it filters through data efficiently and preserves the features and timing of each graph. Understanding and testing these different filters provides knowledge about the applications, benefits, and consequences of the various types of filters and reveals new types of filters that can be tested on data for effectiveness.

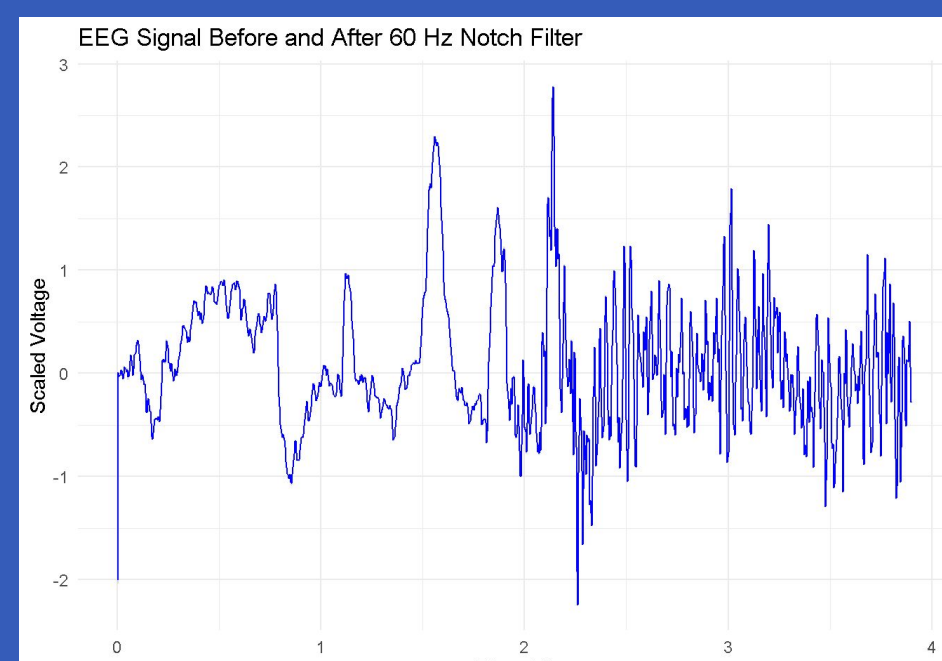
Background Information

- Filtration of EEG data is crucial since most biosignals are incredibly weak and are frequently obscured
 - Random noise, power lines, psychological interference, motion artifacts, and incorrect electrode placements are examples of noise types that can interfere with signal detection (Pant et. al 2024)
- Filtering can also assist with seizure detection to prevent the number of false alarms in tracking methods (Ingolfsson et. al 2024)
 - Filtration makes it easier to manually locate EEG signals when seizures occur (Shoka et. al 2023)
- The prevalence of seizures and epilepsy is growing rapidly, especially after the COVID-19 pandemic (Sen 2022)

Figure 1: Comparison of brain signals before and after the 60 Hz notch signal is removed.



Before



After

Methods and Materials

- **RStudio (v.4.5.1)**
 - Extrapolated data from EDF packages in R Studio
 - Packages used: signal, edf, edfReader, dplyr, ggplot2, kza
- Generated a sample synthetic 5 Hz sine wave and 60 Hz sine wave and applied different filters to display time and frequency-domain plots
 - **Butterworth**: applied zero-phase Butterworth bandpass filters with filter orders of 2, 4, and 8 to test whether changing the order would significantly distort the data
 - **Passband**: designed to emphasize frequencies between 1 and 40 Hz since it covers the brain's major frequencies
 - **Stopband**: frequencies below 1 Hz and above 40 Hz are attenuated (emphasized)
 - **Filter Order**: higher orders create a steeper cutoff between passband and stopband, though very high orders might cause ringing (an undesirable oscillation in the time-domain plot)
 - **Moving Average Filter**: smooths the data but does not provide a precise frequency cutoff
 - **Zero-Phase Filter**: smooths the noise without shifting the step to preserve original timing

Results

- **Butterworth Bandpass Filter**
 - This filter was effective to isolate the specific Hz that were needed to filter out noise
 - Preferable for real-life applications as opposed to full data-sets (casual filter)
 - It introduces a time-delay which distorts the signal, especially with higher order graphs
- **Moving Average Filter**
 - useful for smoothing random noise while maintaining rising and falling edges in data
 - simple to implement and efficient computationally
 - struggle with preserving signal integrity and can cause time delays
- **Zero-Phase Filter**
 - Doesn't have any distortion of the graph or time-delay
 - Preserves the timing of the features of the graphs
 - Creates a sharper transition band
 - Non-casual: entire signals are needed to record the process and cannot be used for real-life applications

The Butterworth bandpass is most useful for real-life applications where EEG data is being analyzed simultaneously with detection and to make urgent diagnoses. The zero-phase filter is preferred when analyzing entire signals. It preserves the features and timing of the graphs and has accurate transitions.

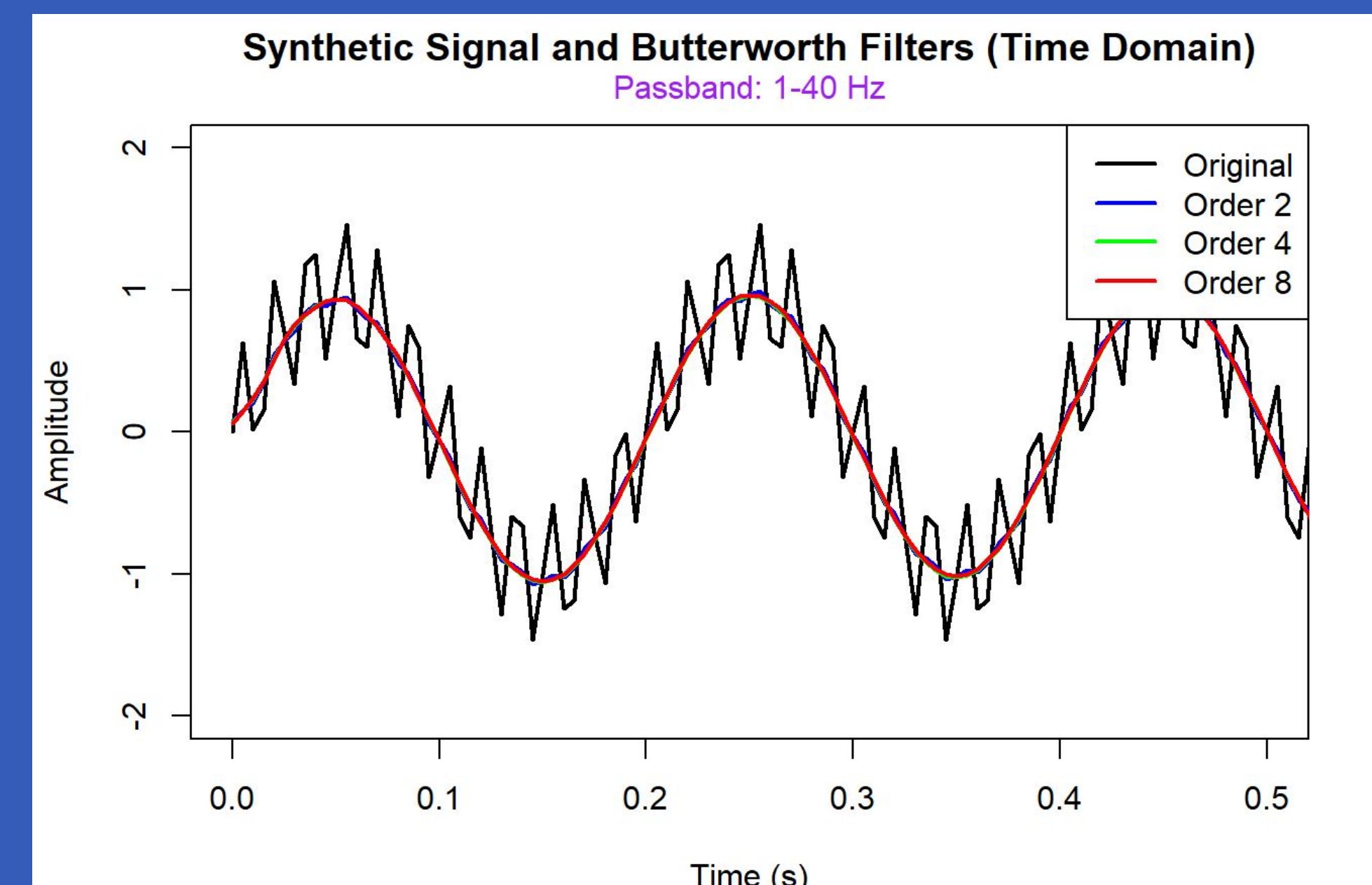


Figure 2: Graph with the original synthetic signal and the smooth curve after the Butterworth Filter is tested at three different orders.

Butterworth Bandpass vs. Moving Average Filtering
Butterworth Passband: 1-40 Hz

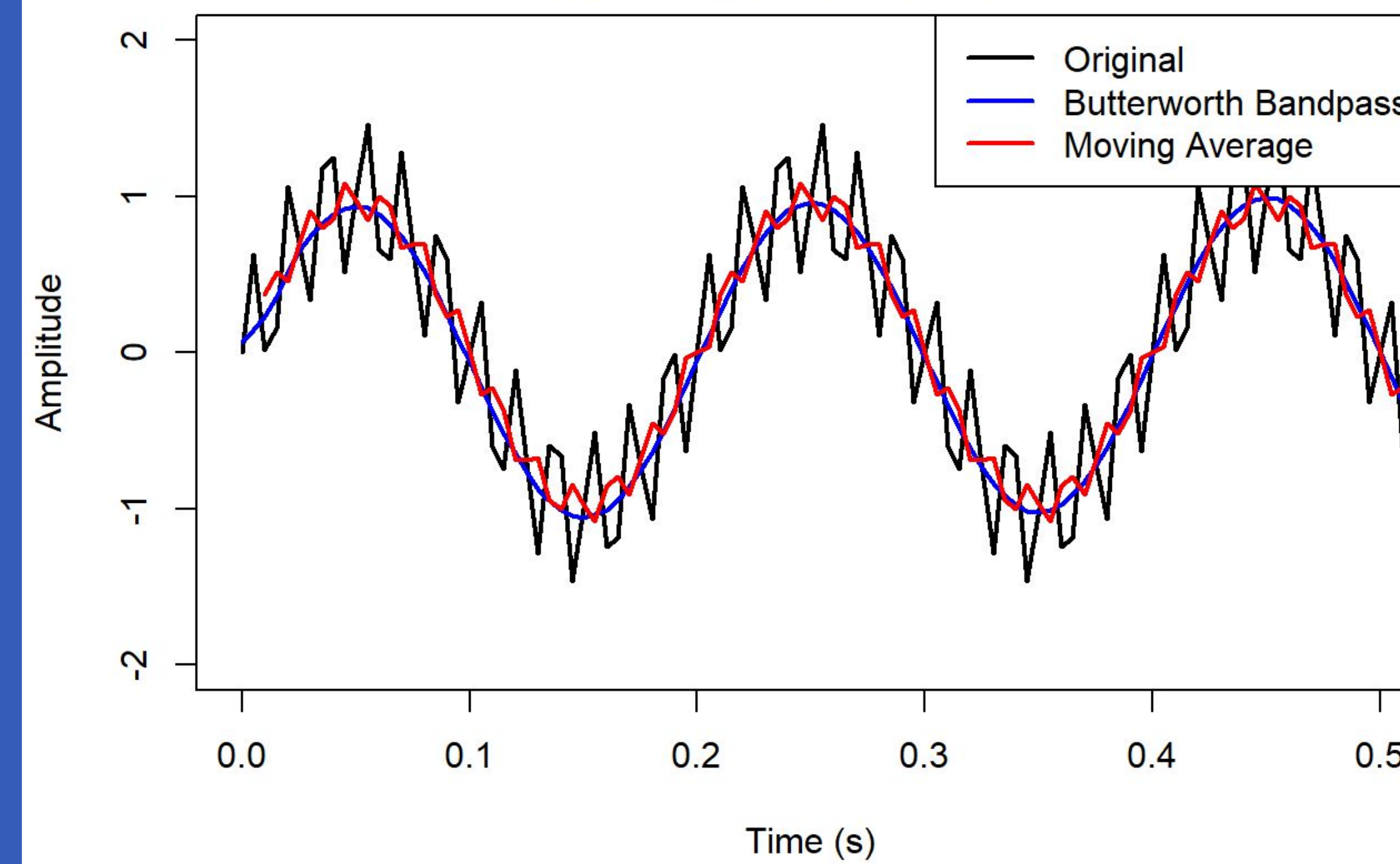


Figure 3: Graph comparing the effectiveness of the Butterworth bandpass with the moving average filter.

The 10-20 System

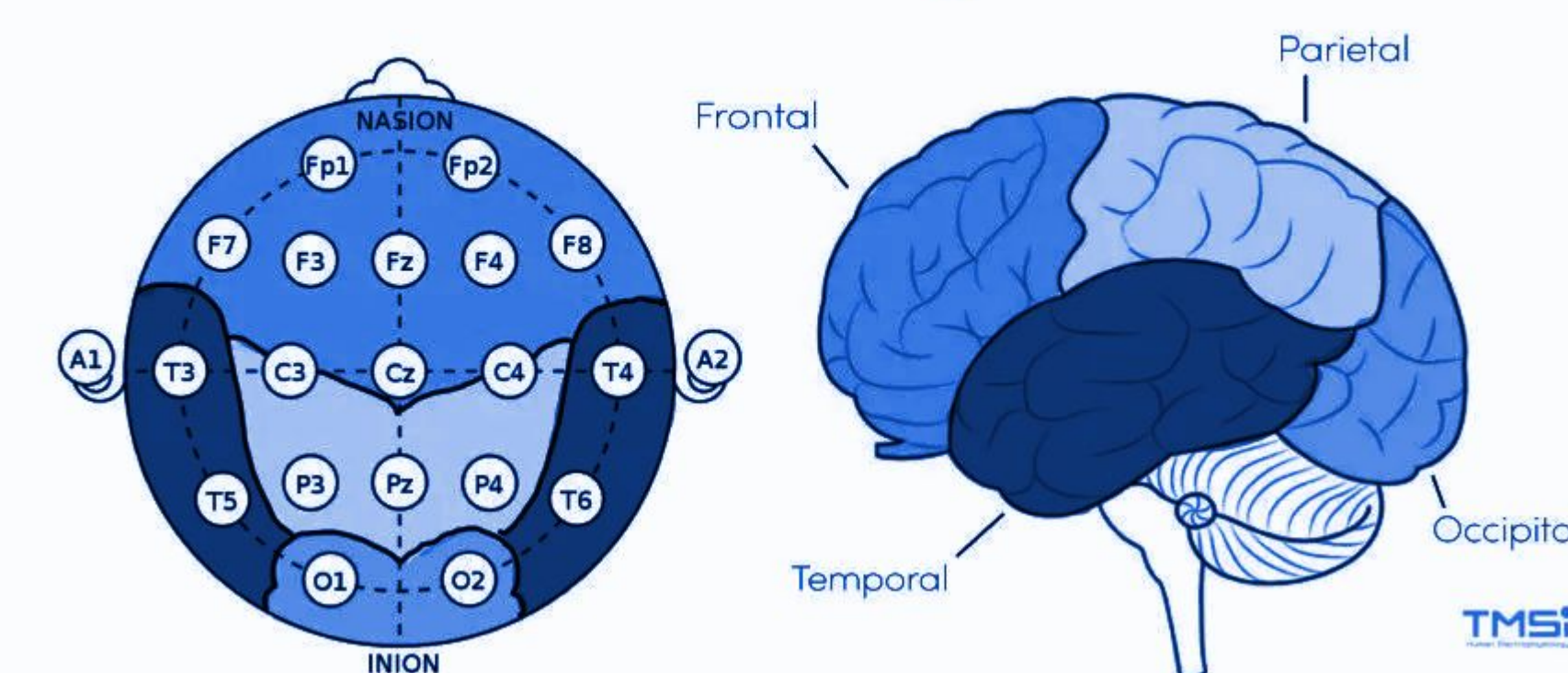


Figure 4: This representation of the brain demonstrates how different areas of the brain correlate with certain signals can be used to identify where mental abnormalities originate from.

Conventional vs. Zero-Phase Filtering

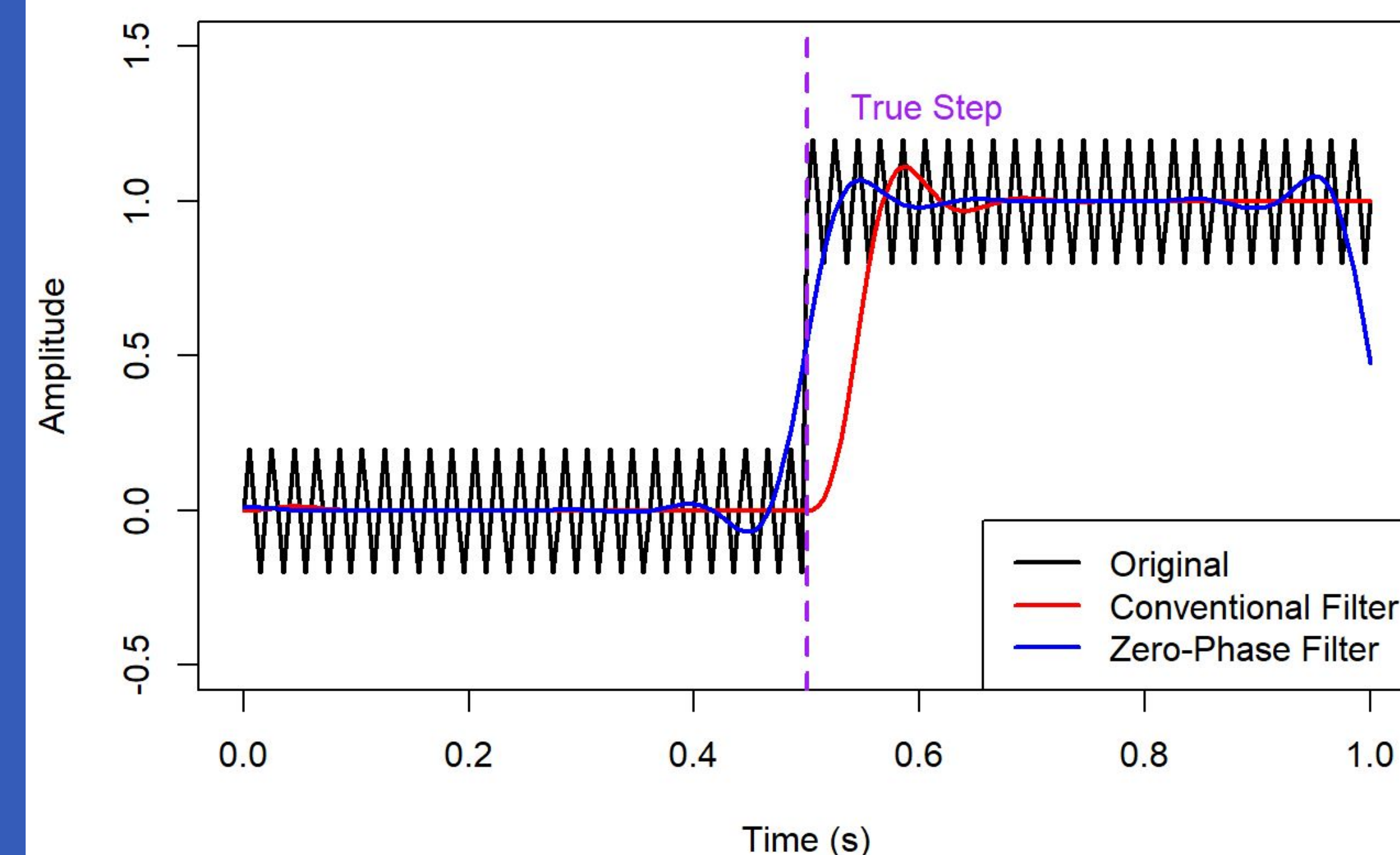


Figure 5: Graph comparing the accuracy of conventional (moving average) filter in comparison to the zero-phase filter by measuring change in filters to true step in data.

Discussion

- With the exception of filtration, automated preprocessing methods tend to be ineffective in removing noise from EEGs, but are necessary to process massive datasets
 - Improving filtration increases the efficiency and accuracy of analysis
- Effective signal filtration is critical in wearable IoT devices for seizure detection for epilepsy patients
 - Improving signal filtration through the Butterworth bandpass, which can be used in real-life applications, can be used in IoT devices to anticipate seizures when neurons start to flare in seizure patterns
- Other potential filtration methods could be low-pass or high-pass filters using the Butterworth filters or a finite impulse response filter
- In zero-phase filters, the effects of symmetric smearing of the signal need to be tested to preserve the delay in the time-domain
 - Symmetric smearing is where a signal is blurred or averaged in a way that affects both sides of a given point equally
 - This could lead to slight distortions in data

Conclusion

- For applications in seizure detection for epilepsy patients, a **Butterworth bandpass** filter is the most effective method for filtration
- To remove noise from large datasets, a **zero-phase filter** filters out noise while maintaining accurate changes in data in relation to time
- Data filtration allows researchers and doctors to determine the location where a seizure began, to detect patterns and prevent future episodes
- Future experimentation could include testing different orders in the Butterworth bandpass filter, low-pass or high-pass filters using the Butterworth filters, or a finite impulse response filter

Acknowledgements

I would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University for providing me with the necessary tools to conduct this research.

References

- Close B, Zurbenko I, Sun M (2020). „kza: Kolmogorov-Zurbenko Adaptive Filters“. doi:10.32614/CRAN.package.kza <https://doi.org/10.32614/CRAN.package.kza>, R package version 4.1.0.1, <https://CRAN.R-project.org/package=kza>
- Delorme, A. (2023). EEG is better left alone. Scientific Reports, 13(1). <https://doi.org/10.1038/s41598-023-27528-0> eegfilter function - RDocumentation. (n.d.). <https://www.rdocumentation.org/packages/eegkit/versions/1.0-5/topics/eegfilter>
- Goudrain, E., Grimaud, N., Healy, E. W., & Béra, J. (2007). Effect of spectral smearing on the perceptual segregation of vowel sequences. Hearing Research, 231(1-2), 32-41. <https://doi.org/10.1016/j.heares.2007.05.001>
- H. Wickham. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York, 2016.
- Henelius A (2016). „edf: Read Data from European Data Format (EDF and EDF+) Files“. doi:10.32614/CRAN.package.edf <https://doi.org/10.32614/CRAN.package.edf>, R package version 1.0.0. <https://CRAN.R-project.org/package=edf>
- Höllwieser, R., & Engelhardt, M. (2015). Approaching SU(2) gauge dynamics with smeared Z(2) vortices. Physical Review D: Particles, Fields, Gravitation, and Cosmology/Physical Review D: Particles, Fields, Gravitation, and Cosmology, 92(3). <https://doi.org/10.1103/physrevd.92.034502>
- Ingolfsson, T. M., Benatti, S., Wang, X., Bernini, A., Ducauret, P., Ryvlin, P., Beniczky, S., Benini, L., & Cossetini, A. (2024). Minimizing artifact-induced false alarms for seizure detection in wearable EEG devices with gradient-boosted tree classifiers. Scientific Reports, 14(1). <https://doi.org/10.1038/s41598-024-52551-0>
- Pant, A., Kumar, A., Verma, C., & Illés, Z. (2024). Comparative exploration on EEG signal filtering using window control methods. Results in Control and Optimization, 100485. <https://doi.org/10.1016/j.rico.2024.100485>
- Sen, A. The risk of seizures and epilepsy is higher after COVID than after the flu. (2022, November 21). Haiku. <https://www.psych.ox.ac.uk/news/the-risk-of-seizures-and-epilepsy-is-higher-after-covid-than-after-the-flu>
- signal developers (2023). „signal: Signal processing“. <https://r-forge.r-project.org/projects/signal/>
- Shoka, A. a. E., Dessouky, M. M., El-Sayed, A., & Hermand, E. E. (2023). EEG seizure detection: concepts, techniques, challenges, and future trends. Multimedia Tools and Applications, 82(27), 42021-42051. <https://doi.org/10.1007/s11042-023-15052-2>
- Systems, A. M. (2025, March 12). TMSI — an Artinis company — What is the 10-20 System for EEG? TMSI — an Artinis Company. <https://www.tmsi-artinis.com/blog/the-10-20-system-for-eeeg>
- Vis J (2019). „edfReader: Reading EDF(+) and BDF(+) Files“. doi:10.32614/CRAN.package.edfReader <https://doi.org/10.32614/CRAN.package.edfReader>, R package version 1.2.1, <https://CRAN.R-project.org/package=edfReader>
- Wickham H, François R, Henry L, Müller K, Vaughan D (2023). „dplyr: A Grammar of Data Manipulation“. doi:10.32614/CRAN.package.dplyr <https://doi.org/10.32614/CRAN.package.dplyr>, R package version 1.1.4, <https://CRAN.R-project.org/package=dplyr>
- Zetadmin. (2024, February 11). How digital EEG filters impact EEG signal morphology. Zeto EEG. <https://zeto-inc.com/blog/eeg-signal-enhancement-digital-eeeg-filters/>